```
> restart;#The Phonon Planck's "constant"
> Digits:=20;
```

$$Digits := 20 \tag{1}$$

```
> E(nu):=Em+E0*ln(1+2*Tq1*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(2\, Tq1\, A\, \nu + 1) + C\, \nu^2 \tag{2}$$

```
> c:=299792458;nu1:=435;ec:= 1.602176634*10^(-19);E1:=3.259;T:=297;
  va:=343;Tq1:=c/va*1.765*10^(-19);ha:=0.007486790719; ha*ec;Tq1*
  105.2757803;2*Tq1*32/3*Pi^2;
```

$$c := 299792458$$

$$v1 := 435$$

$$ec := 1.6021766340000000000 \times 10^{-19}$$

$$E1 := 3.259$$

$$T := 297$$

$$va := 343$$

$$Tq1 := 1.5426638144897959184 \times 10^{-13}$$

$$ha := 0.007486790719$$

$$1.1995161153629859846 \times 10^{-21}$$

$$1.6240513681098771170 \times 10^{-11}$$

$$3.2481027355498213704 \times 10^{-11} \tag{3}$$

```
> E(nu):=Em+E0*ln(1+A*3.2481027355498213704*10^(-11)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 3.2481027355498213704 \times 10^{-11}\, A\, \nu) + C\, \nu^2 \tag{4}$$

```
> E1:=subs(nu=435,E(nu));
```

$$E1 := Em + E0 \ln(1 + 1.4129246899641722961 \times 10^{-8}\, A) + 189225\, C \tag{5}$$

```
> E2:=subs(nu=528,E(nu));
```

$$E2 := Em + E0 \ln(1 + 1.7149982443703056836 \times 10^{-8}\, A) + 278784\, C \tag{6}$$

```
> E3:=subs(nu=25000,E(nu));
```

$$E3 := Em + E0 \ln(1 + 8.1202568388745534260 \times 10^{-7}\, A) + 625000000\, C \tag{7}$$

```
> solve({E1=3.259,E2=3.95});
```

$$\{A = A,\ C = 0.00001116582364698131957 7\, E0 \ln(1. + 1.4129246899641722961 \times 10^{-8}\, A) - 0.00001116582364698131957 7\, E0 \ln(1. + 1.7149982443703056836 \times 10^{-8}\, A) + 7.7155841400640918277 \times 10^{-6},\ E0 = E0,\ Em = 2.1128529796000401970\, E0 \ln(1. + 1.7149982443703056836 \times 10^{-8}\, A) - 3.1128529796000401970\, E0 \ln(1. + 1.4129246899641722961 \times 10^{-8}\, A) + 1.7990185910963722239\} \tag{8}$$

```
> sol:=solve({E1=3.259,E2=3.95,E3=0},{Em,E0,C,A});
```

$$sol := \{A = A,\ C = -(0.0010000000000000000000\ (-3259.\ln(1. + 1.7149982443703056836 \times 10^{-8} A) - 691.\ln(1. + 8.1202568388745534260 \times 10^{-7} A) + 3950.\ln(1. + 1.4129246899641722961 \times 10^{-8} A)))/(89559.\ln(1. + 8.1202568388745534260 \times 10^{-7} A) - 6.24810775 \times 10^{8}\ln(1. + 1.7149982443703056836 \times 10^{-8} A) + 6.24721216 \times 10^{8}\ln(1. + 1.4129246899641722961 \times 10^{-8} A)),\ E0 = -(4.3203611830600000000 \times 10^{8})/(89559.\ln(1. + 8.1202568388745534260 \times 10^{-7} A) - 6.24810775 \times 10^{8}\ln(1. + 1.7149982443703056836 \times 10^{-8} A) + 6.24721216 \times 10^{8}\ln(1. + 1.4129246899641722961 \times 10^{-8} A)),\ Em = (0.0020000000000000000000\ (8.0559153 \times 10^{7}\ln(1. + 8.1202568388745534260 \times 10^{-7} A) - 1.018437500000 \times 10^{12}\ln(1. + 1.7149982443703056836 \times 10^{-8} A) + 1.234375000000 \times 10^{12}\ln(1. + 1.4129246899641722961 \times 10^{-8} A)))/(89559.\ln(1. + 8.1202568388745534260 \times 10^{-7} A) - 6.24810775 \times 10^{8}\ln(1. + 1.7149982443703056836 \times 10^{-8} A) + 6.24721216 \times 10^{8}\ln(1. + 1.4129246899641722961 \times 10^{-8} A))\} \quad (9)$$

```
> A:=9.949747*10^6;evalf(sol[4]);#Em
```

$$A := 9.949747000000 \times 10^{6}$$

$$Em = -0.26049922890091899376 \quad (10)$$

```
> Em := -0.26049922890091899376;Em;
```

$$Em := -0.26219418286029721944$$

$$-0.26219418286029721944 \quad (11)$$

```
> evalf(sol[3]);#E0
```

$$E0 = 26.892256305433408378 \quad (12)$$

```
> E0 := 26.892256305433408378;E0;
```

$$E0 := 26.914388251007721676$$

$$26.914388251007721676 \quad (13)$$

```
> evalf(sol[2]);;#C
```

(14)

$$C = -9.4502695574974682733 \times 10^{-8} \quad (14)$$

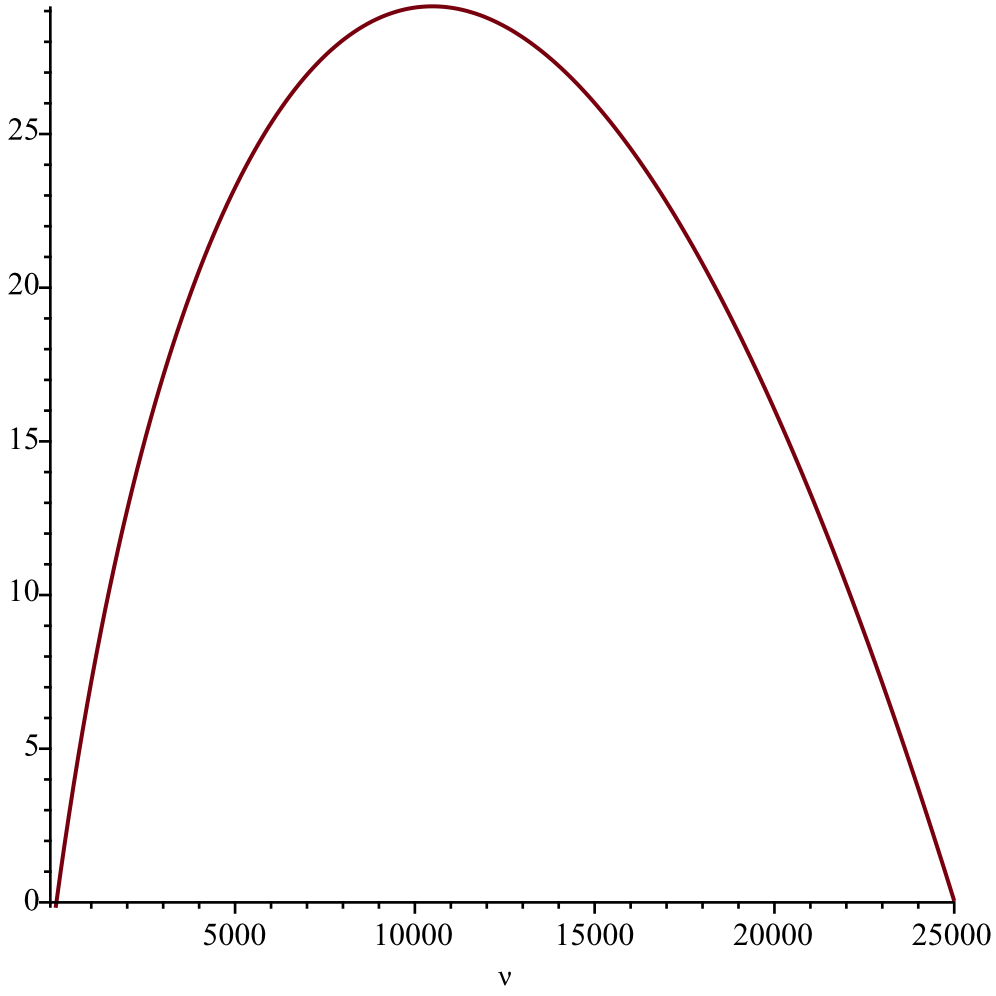
```
> C:=-9.4502695574974682733*10^(-8);
```

$$C := -9.4502695574974682733 \times 10^{-8} \quad (15)$$

```
> E(nu):=Em+E0*ln(1+A*3.2481027355498213704*10^(-11)*nu)+ C*nu^2;
```

$$E(\nu) := -0.26219418286029721944 + 26.914388251007721676 \ln(1 + 0.00032317800448728628531\,\nu) - 9.4502695574974682733 \times 10^{-8}\,\nu^2 \quad (16)$$

```
> plot(E(nu),nu=10..25000);
```




```
> evalf(subs(nu=435,E(nu)));evalf(subs(nu=528,E(nu)));evalf(subs
  (nu=480,E(nu)));
```

$$3.2602162608519398957$$

$$3.9517919094482611277$$

$$3.5973279580401586754 \quad (17)$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=30,E(nu)));evalf(subs(nu=
  13350,E(nu)));#The max value is ~26.7eV at frequency ~13350Hz
```

```
[30Hz÷24khz]
```

$$-0.26219418286029721944$$
$$-0.00259194059505890622$$
$$27.853768233661099002 \quad \textbf{(18)}$$